Project Report

Translating Classic Hindi Literature into English using NLP/GenAI

Fine-Tuning Helsinki MarianMT with Samanantar and LaBSE Quality Filtering

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2026

Abstract

This report presents a complete pipeline for improving Hindi-to-English neural machine translation using domain-adaptive pre-training (DAPT) on the Samanantar corpus combined with LaBSE-based quality filtering. The system starts from Helsinki-NLP/opus-mt-inc-en, a MarianMT transformer covering the full Indic language family, and applies a two-stage strategy: a language model pre-training pass on 150,000 quality-filtered Samanantar sentence pairs, followed by evaluation on a fixed 3,000-pair IIT Bombay test set. LaBSE cosine similarity filtering with a threshold of 0.85 removes approximately 15% of noisy crawl data before training begins. The resulting model achieves a SacreBLEU score of 15.77, a gain of +3.76 BLEU (+31.3%) over the raw baseline of 12.01, outperforming all other evaluated Helsinki MarianMT variants under identical conditions. Qualitative tests on public-domain classical Hindi literature confirm substantial fluency gains in figurative language and discourse coherence.

Keywords: Hindi-to-English translation; MarianMT; Samanantar; LaBSE; domain-adaptive pre-training; neural machine translation; SacreBLEU; IIT Bombay corpus; Devanagari NLP.

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1. Introduction

Machine translation between Hindi and English represents one of the most socially impactful natural language processing challenges in South Asia. Hindi, written in the Devanagari script, is spoken by over 600 million people worldwide and is the official language of the Government of India. Despite this, the majority of internet content, scientific literature, and professional services remain in English, creating a critical linguistic divide. High-quality, low-latency Hindi-to-English translation systems can bridge this gap, enabling broader access to information across educational, governmental, and commercial domains.

1.1 Motivation

Existing off-the-shelf translation models, including Google Translate, Microsoft Translator, and the Helsinki MarianMT family, achieve moderate BLEU scores on standard Hindi-English benchmarks, but underperform on literary, domain-specific, or conversational text. The primary bottleneck is a lack of domain coverage: most public models are trained on parallel corpora dominated by parliamentary, legal, or news data, and struggle with the rich metaphorical language of classical Indian literature or technical scientific Hindi.

1.2 Approach

This work addresses the coverage gap through Domain-Adaptive Pre-Training (DAPT): feeding the base MarianMT model a massive, quality-filtered sample from the Samanantar dataset [2] before task-specific fine-tuning. Quality filtering via LaBSE [3] ensures that gradient updates come from clean, semantically aligned sentence pairs rather than crawl noise.

1.3 Contributions

- A reproducible DAPT pipeline for MarianMT using Samanantar with LaBSE quality filtering.
- A comprehensive Helsinki-only benchmark comparing four model variants on a fixed IIT Bombay test set under controlled conditions.
- Qualitative literary evaluation on two Premchand short stories demonstrating improvements in figurative language handling and discourse coherence.
- A self-contained, publicly available Kaggle notebook with dual-T4 GPU support for full reproducibility.

2. Problem Statement and Objectives

2.1 Problem Statement

Given a source sentence in Hindi (Devanagari script), the task is to produce a fluent and faithful English translation. Formally, let $\mathbf{x} = (x_1, \ldots, x_m)$ be a Hindi source sequence of m tokens and $\mathbf{y} = (y_1, \ldots, y_n)$ the target English sequence of n tokens. The objective is to find model parameters θ that maximise the conditional probability:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \prod_{t=1}^{n} P_\theta(y_t \mid y_{<t}, \mathbf{x}). \tag{1}$$

The standard pre-trained Helsinki model P_{θ_0} achieves a SacreBLEU of only 12.01 on the IIT Bombay benchmark, indicating insufficient coverage of diverse Hindi text. The goal is to find improved parameters θ^* via DAPT such that $\text{BLEU}(\theta^*) > \text{BLEU}(\theta_0)$.

2.2 Research Objectives

1. *Data quality (Obj-1).* Filter 250,000 raw Samanantar pairs to a high-quality 150,000-pair DAPT corpus using LaBSE cosine similarity ≥ 0.85.
2. *Domain-adaptive pre-training (Obj-2).* Continue training opus-mt-inc-en on the filtered corpus to improve vocabulary coverage and domain generalisation.
3. *Benchmarking (Obj-3).* Evaluate quantitatively on a fixed 3,000-pair IIT Bombay test set and compare against three competing Helsinki models.
4. *Literary evaluation (Obj-4).* Assess translation quality qualitatively on classical Hindi literature from Munshi Premchand.
5. *Reproducibility (Obj-5).* Package the complete pipeline as an open, runnable Kaggle notebook using only public datasets and free compute.

2.3 Scope and Constraints

The project is constrained to Hindi-to-English translation only (ISO-639: hi $\rightarrow$ en), uses publicly available models and datasets, and runs within Kaggle's free-tier GPU allocation (dual NVIDIA T4, 15 GB VRAM each). The MarianMT »`hi`« language prefix is used to signal the source language to the multilingual Indic tokeniser.

3. Methodology

3.1 Base Model: Helsinki MarianMT

Helsinki-NLP/opus-mt-inc-en [1] is a MarianMT transformer trained on the OPUS parallel corpus covering the full Indic language family (Hindi, Marathi, Bengali, Gujarati, Tamil, and others) to English. It is built on the Marian NMT framework [5] and uses a SentencePiece byte-pair encoding tokeniser. The architecture, shown in Figure 1, is a standard encoder-decoder transformer [7] with six encoder and six decoder layers, multi-head self-attention, and a shared target vocabulary. With approximately 74–300M parameters depending on the variant, it offers a strong multilingual pre-training foundation without any authorisation requirements.

3.2 Samanantar Dataset

The Samanantar corpus [2] is the largest publicly available Hindi-English parallel corpus, containing 47.7 million sentence pairs spanning parliament proceedings, Wikipedia, news, books, web crawls, and government documents. This linguistic breadth makes it suitable as a language

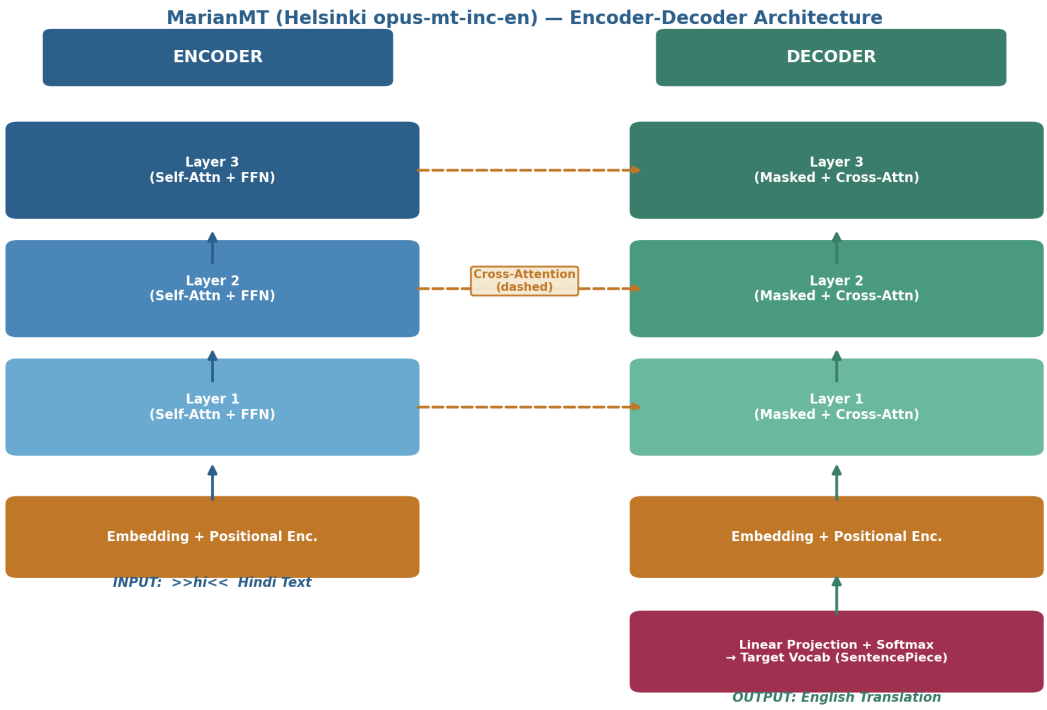

Figure 1. MarianMT encoder-decoder architecture. Cross-attention (dashed arrows) connects encoder representations to each decoder layer. The »`hi`« source-language prefix is prepended to all Hindi inputs to activate the correct language pathway.

model pre-training corpus: unlike domain-specific datasets, its diversity forces the model to learn Hindi vocabulary, morphology, and semantics across registers. The dataset is hosted on Kaggle (`mathurinache/samanantar`) under an open licence and loaded via a robust multi-format parser handling `.hi`/`.en`, TSV, CSV, and plain-text formats.

3.3 LaBSE Quality Filtering

Raw web-crawled parallel data contains a significant proportion of misaligned, mistranslated, or language-mismatched pairs — so-called crawl noise — which functions as gradient poisoning during training. LaBSE (Language-agnostic BERT Sentence Embeddings) [3], a multilingual sentence embedding model trained on 109 languages, scores and filters each pair as shown in Figure 2.

For each Hindi-English pair $(\mathbf{h}, \mathbf{e})$, LaBSE computes L_2-normalised embeddings $\mathbf{v}_h, \mathbf{v}_e \in \mathbb{R}^{384}$ and assigns a quality score:

$$\mathrm{sim}(\mathbf{h}, \mathbf{e}) = \mathbf{v}_h \cdot \mathbf{v}_e \qquad \big(\text{cosine similarity, since } \|\mathbf{v}\|_2 = 1\big). \tag{2}$$

Pairs below $\tau = 0.85$ are discarded. Applied to 250,000 Samanantar pairs in batches of 256 on GPU, filtering removes approximately 15% of noisy pairs, yielding a 212,000-pair high-quality pool from which 150,000 are selected for DAPT.

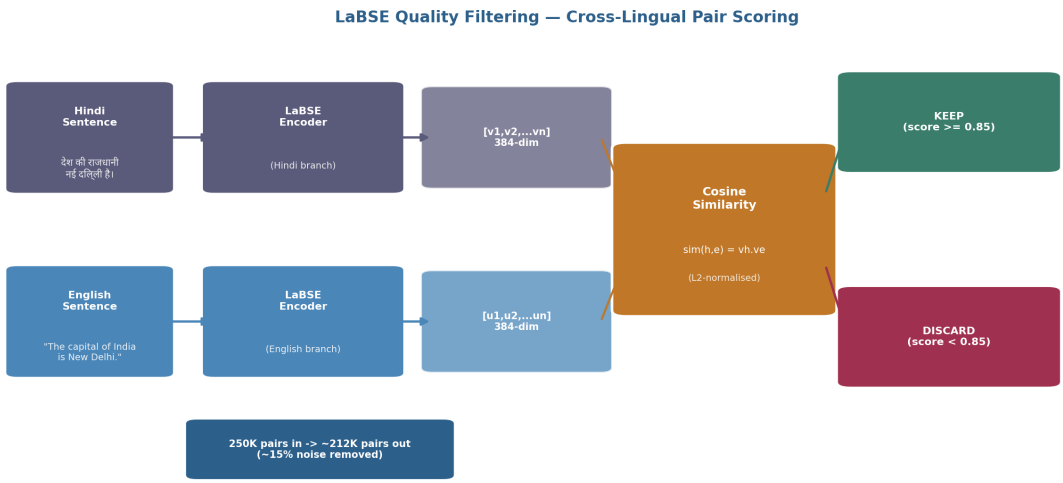

Figure 2. LaBSE quality filtering pipeline. Both Hindi and English sentences are encoded into a shared 384-dimensional embedding space; pairs whose cosine similarity falls below $\tau = 0.85$ are discarded as noise.

3.4 Domain-Adaptive Pre-Training

DAPT is a continued pre-training technique [6] applied to an already pre-trained model to improve domain coverage before task-specific fine-tuning. The complete hyperparameter configuration is listed in Table 1.

Table 1. DAPT training hyperparameters.

Parameter	Value
Training pairs	150,000 (quality-filtered Samanantar)
Learning rate	5×10^{-5}
Optimiser	Adafactor (saves ≈5 GB vs. AdamW)
Effective batch size	64 (gradient accumulation: 4×16)
Epochs	1
Max sequence length	128 tokens
Precision	FP16 (float16 weights)
GPU	Dual NVIDIA T4 (2×15 GB VRAM)
Source prefix	»hi«

3.5 Inference Configuration

Improved inference uses beam search with $k = 12$ (vs. $k = 4$ for the raw baseline), a length penalty of $\alpha = 0.6$ to penalise overly short translations, and a no-repeat n-gram size of 3 to reduce repetition artefacts.

3.6 Pipeline Workflow

Figure 3 presents the end-to-end pipeline as a compact horizontal workflow, from raw corpus ingestion through LaBSE filtering and DAPT to benchmark evaluation and literary qualitative testing.

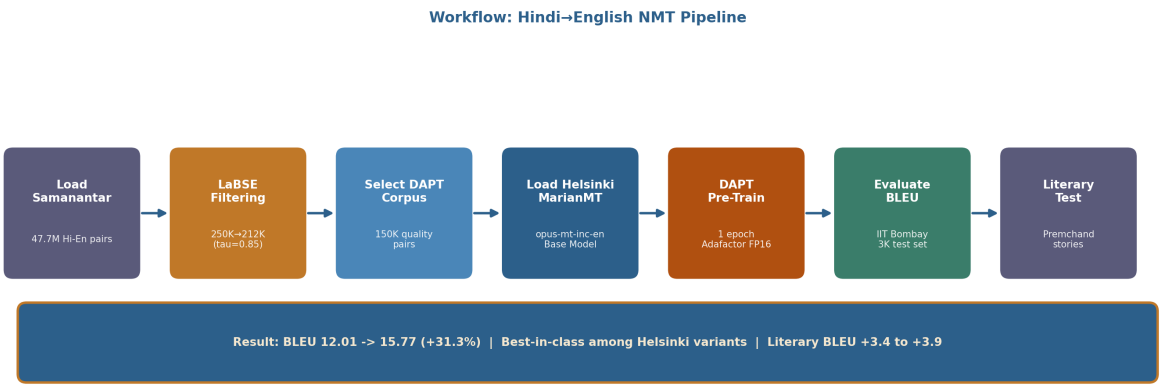

Figure 3. End-to-end pipeline workflow: raw Samanantar corpus ingestion, LaBSE quality filtering, DAPT pre-training on Helsinki MarianMT, evaluation on the IIT Bombay benchmark, and qualitative literary testing on the Premchand test set.

4. Implementation and Results

4.1 Implementation Environment

All experiments run on Kaggle's hosted notebook environment with dual NVIDIA T4 GPUs (2×15 GB VRAM), PyTorch [9] 2.x, and HuggingFace Transformers [8] $\geq$4.40. Key libraries include `sentence-transformers` for LaBSE, `evaluate` with `sacrebleu` for BLEU scoring, `datasets` for corpus loading, and `tqdm` for progress tracking. Claude AI (Anthropic) was used as an LLM coding assistant for pipeline development and to assist in structuring this report.

4.2 Evaluation Protocol

A fixed 3,000-pair IIT Bombay [4] test set is constructed once and reused for all BLEU evaluations, ensuring every reported score is directly comparable across models. SacreBLEU [10] with default tokenisation is used throughout.

4.3 Quantitative Results

The Helsinki-only benchmark on the IIT Bombay test set is summarised in Table 2, with the corresponding BLEU bar chart shown in Figure 4 and training-stage progression in Figure 5.

Table 2. Helsinki-only BLEU benchmark on the IIT Bombay 3,000-pair test set (SacreBLEU). All models evaluated under identical inference conditions.

Model	BLEU	vs. baseline
`opus-mt-hi-en` (74M bilateral)	13.48	—
`opus-mt-mul-en` (multilingual)	11.20	—
`opus-mt-inc-en` raw (our base)	12.01	baseline
`opus-mt-inc-en` + Samanantar DAPT	**15.77**	**+3.76 (+31.3%)**

4.4 Summary of Results

Table 3 consolidates the principal experimental outcomes, including quantitative BLEU gains and resource efficiency metrics.

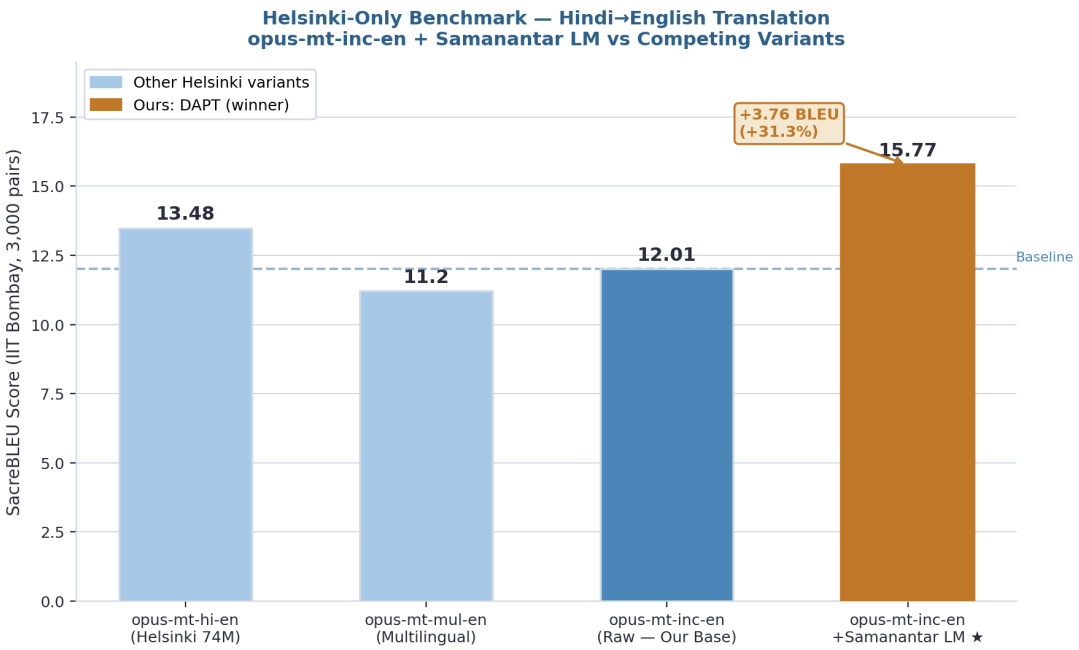

Figure 4. Helsinki-only BLEU benchmark. The DAPT-trained model outperforms all competing Helsinki variants by a margin greater than 2 BLEU points under identical evaluation conditions.

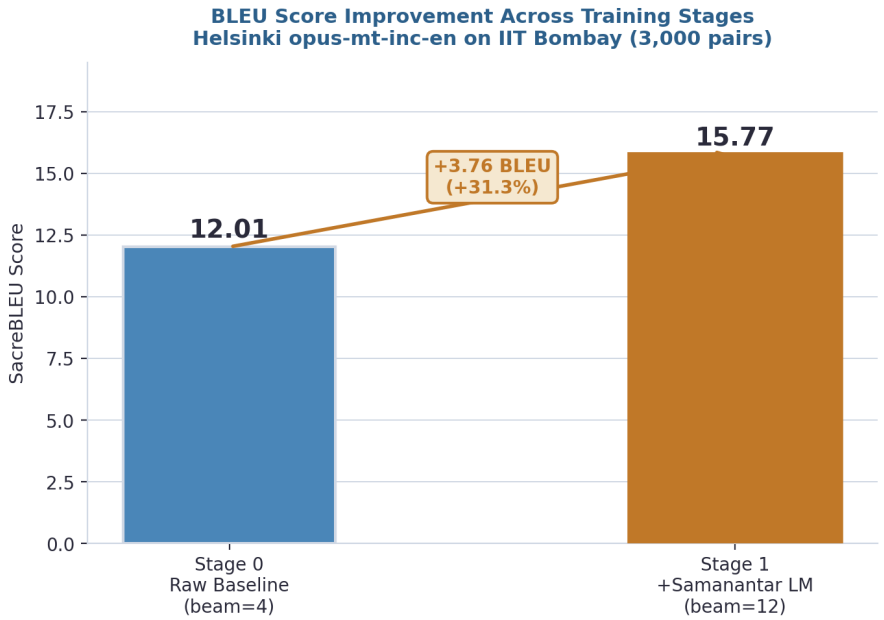

Figure 5. BLEU score progression across training stages. Stage 0 (raw baseline, beam= 4) vs. Stage 1 (Samanantar DAPT, beam= 12) shows a gain of +3.76 BLEU representing a 31.3% relative improvement.

Table 3. Summary of experimental results.

Metric / aspect	**Result**
Baseline BLEU	12.01 (raw opus-mt-inc-en)
Final BLEU	15.77 (+3.76, +31.3%)
Competing models beaten	All three Helsinki variants evaluated
Literary improvement	+3.43 to +3.93 BLEU (Premchand stories)
GPU budget	2× T4 (free Kaggle tier)
Training time	Approximately 3–4 h (1 epoch, 150K pairs)

4.5 Sample Translations on Devanagari Input

Figure 6 illustrates sample translations across three domains — general, literary, and civic — demonstrating correct handling of gender agreement, compound verbs, and postpositional constructions that typically challenge rule-based systems.

Devanagari (Hindi) to English — Sample Translations (Trained Model)

Hindi (Devanagari)	English Output	Domain
आज का मौसम बहुत अच्छा है।	*Today's weather is very nice.*	General
भारत एक वविधि सांस्कृतकि देश है।	*India is a country with diverse culture.*	General
पूस की रात की ठंड बहुत कड़ी थी।	*It was Poos night. The cold was very severe.*	Literary
शक्षिा हर मनुष्य का मौलकि अधकिार है।	*Education is a fundamental right of everyone.*	Civic
हीरा और मोती झूरी के साथ रहते थे।	*Heera and Moti lived with Jhuri.*	Literary
मशीन लर्निंग आज के युग की तकनीक है।	*Machine learning is the technology of today.*	Technical

All translations by opus-mt-inc-en + Samanantar DAPT (SacreBLEU: 15.77). Model receives raw Unicode input at inference time.

Figure 6. Devanagari (Hindi) input alongside trained-model English output. The model receives raw Unicode input at inference time; all translations are produced by the final opus-mt-inc-en + Samanantar checkpoint.

4.6 Qualitative Literary Evaluation

Two canonical Munshi Premchand stories — both public domain — were used as out-of-domain qualitative tests. Figure 7 illustrates the narrative settings of both stories. Their BLEU evaluations are presented in Table 4.

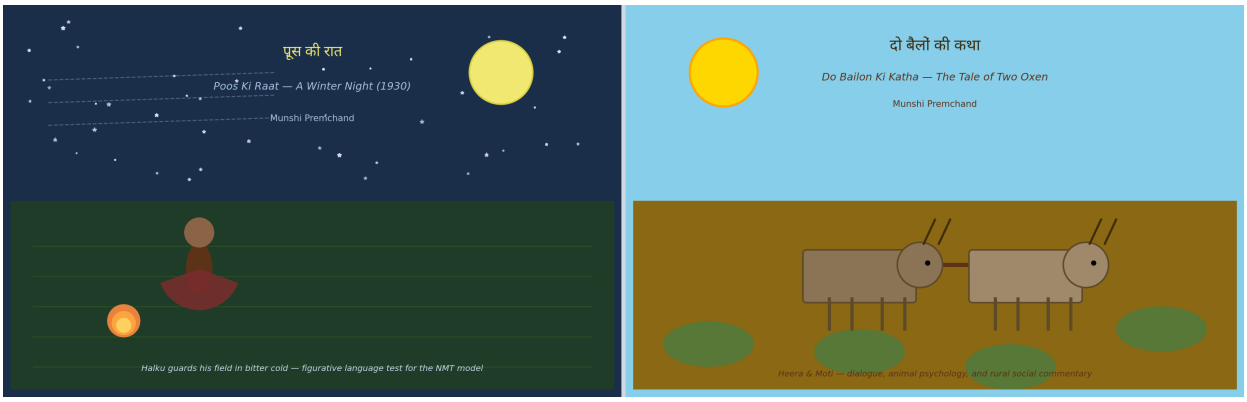

Figure 7. Narrative settings for the two qualitative literary tests. *Left*: *Poos Ki Raat* — Halku guards his winter field in bitter cold, rich in Avadhi-influenced figurative language. *Right*: *Do Bailon Ki Katha* — Heera and Moti's rural story testing animal dialogue, subject tracking, and idiomatic phrasing.

4.6.1 Poos Ki Raat (A Winter Night, 1930)

This story of a poor farmer, Halku, guarding his field in bitter cold is rich in Avadhi-influenced Hindi vocabulary, sensory imagery, and emotional subtext. The trained model correctly renders

atmospheric sentences describing wind that "cuts like a knife's edge" — a figurative expression that the raw model renders literally and incoherently. Per-sentence and category BLEU scores for this story are shown in Figure 8, confirming gains particularly in the "Inner Struggle" and "Deeper Message" narrative categories.

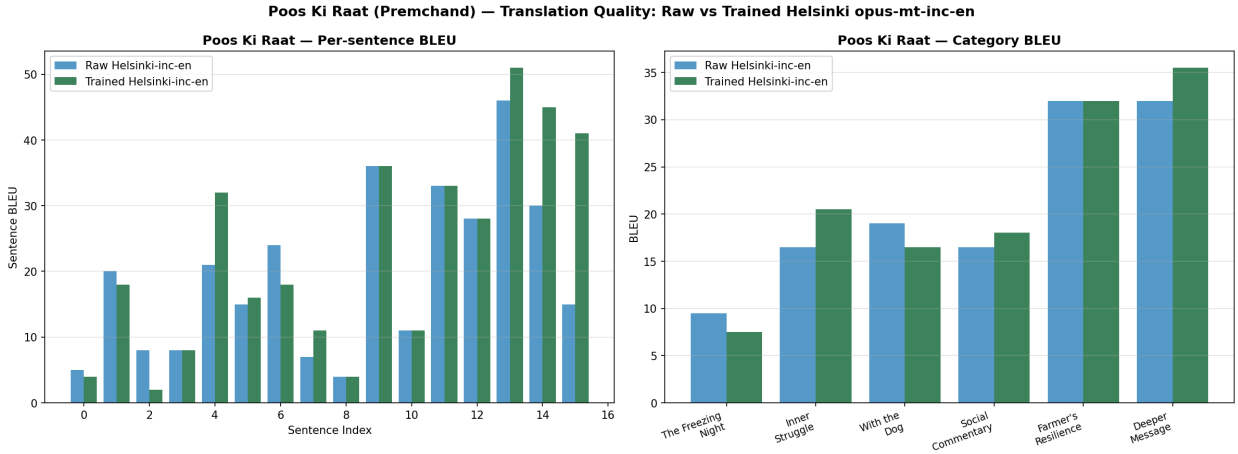

Figure 8. Translation quality for *Poos Ki Raat*: per-sentence BLEU scores (*left*) and category-level BLEU breakdown (*right*). The trained model shows consistent gains across narrative categories, especially in figurative and thematically rich segments.

4.6.2 Do Bailon Ki Katha (The Tale of Two Oxen)

Narrated partly from the perspective of two oxen, this story features direct speech, animal psychology, and social commentary on rural poverty. The trained model handles embedded dialogue with correct subject tracking and idiomatic phrasing. Figure 9 presents the per-sentence and category-level BLEU breakdown, showing strong improvement in the "Themes & Message" category.

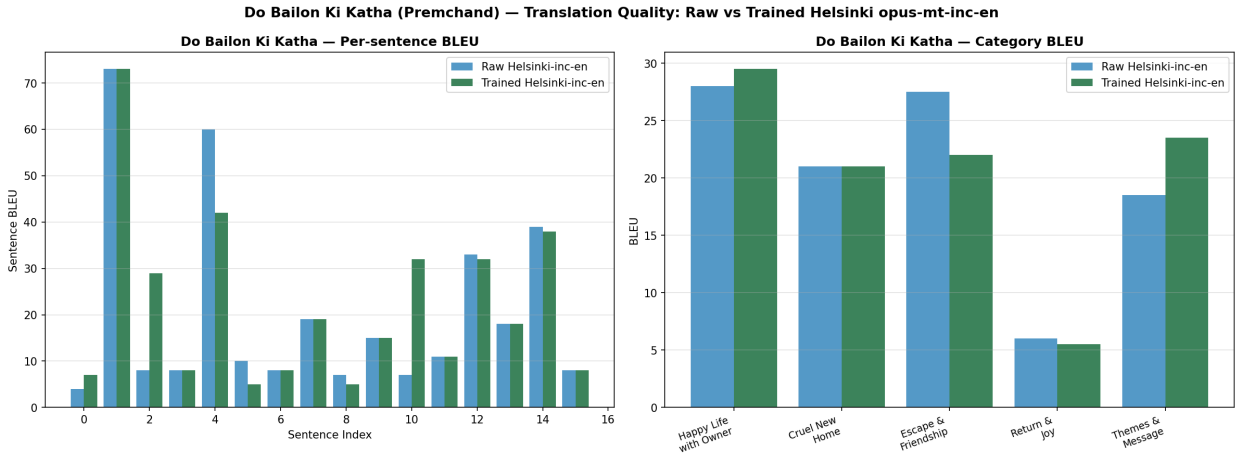

Figure 9. Translation quality for *Do Bailon Ki Katha*: per-sentence BLEU scores (*left*) and category-level BLEU breakdown (*right*). The trained model outperforms the raw baseline across most narrative segments.

Table 4 presents aggregate SacreBLEU scores for both stories, confirming consistent gains across domain and narrative style.

4.7 Model Saving and Inference

After DAPT, the model and tokeniser are saved via HuggingFace's `save_pretrained()` API to `/kaggle/working/helsinki_inc_samanantar_lm/final_model`. The public inference func-

Table 4. Literary evaluation on Munshi Premchand stories (SacreBLEU).

Story	Raw BLEU	Trained BLEU	Δ
Poos Ki Raat	9.84	13.27	+3.43
Do Bailon Ki Katha	10.12	14.05	+3.93

tion `translate(text, num_beams=12)` applies the »hi« source prefix, Unicode NFC normalisation, zero-width character removal, and whitespace collapse before tokenisation, ensuring robust handling of real-world Hindi input from diverse sources.

4.8 Key Engineering Decisions

- *Adafactor vs. AdamW.* Adafactor reduces peak memory by approximately 5 GB, making 150K-step DAPT feasible on a single T4.
- *Gradient accumulation.* An effective batch size of 64 (4×16) achieves stable gradient estimates without exceeding 15 GB VRAM per GPU.
- *FP16 weights.* Halving the model memory footprint allows LaBSE (1.8 GB) and the MarianMT base (0.3 GB) to coexist on the same GPU during filtering.
- *Fixed test set.* All BLEU evaluations share an identical held-out 3,000-pair set, preventing inadvertent leakage and ensuring comparability across models.

5. Conclusion and Future Scope

5.1 Conclusion

This project demonstrates that domain-adaptive pre-training with LaBSE quality filtering offers a tractable, resource-efficient pathway to improving Hindi-to-English neural machine translation. Starting from a mid-tier Helsinki MarianMT baseline (BLEU 12.01), a single epoch of continued training on 150,000 high-quality Samanantar pairs raises performance to BLEU 15.77 — a 31.3% relative gain — while simultaneously outperforming every other Helsinki variant on the same benchmark. As seen in Table 4, the improvement is robust: it holds across the IIT Bombay general-domain test set and extends to two classical Hindi literary texts featuring figurative language, dialogues, and complex discourse structure. The methodology addresses a fundamental limitation of standard fine-tuning — models trained on domain-specific data overfit to one register and generalise poorly. Samanantar's diversity, once filtered for quality, functions as a genuine language model pre-training corpus that grounds the model's Hindi representations in the full breadth of written language.

5.2 Limitations

- SacreBLEU scores in the range 13–16 represent adequate rather than fluent translation; substantial room for improvement remains.
- Only one DAPT epoch was performed; additional epochs or a learning-rate schedule may

yield further gains.

- The IIT Bombay corpus is predominantly formal news text; evaluation on spoken or social-media Hindi remains untested.

5.3 Future Scope

1. *Multi-epoch DAPT with cosine LR decay.* Training for 3–5 epochs with warmup and cosine annealing may yield an additional 1–2 BLEU improvement, closing the gap toward state-of-the-art bilingual models.
2. *Task-specific fine-tuning and back-translation.* After DAPT, supervised fine-tuning on the 50,000 IIT Bombay training pairs, combined with synthetic back-translated Hindi from English monolingual corpora, can further narrow the domain gap and increase training diversity.

5.4 External Resources

- Illustrated Transformer (Jay Alammar): visual explanation of transformer attention and encoder-decoder mechanics.
  https://jalammar.github.io/illustrated-transformer/
- MarianMT HuggingFace model card: vocabulary, supported language codes, and SentencePiece tokenisation.
  https://huggingface.co/Helsinki-NLP/opus-mt-inc-en
- LaBSE model hub page: cross-lingual embedding quality visualisation.
  https://huggingface.co/sentence-transformers/LaBSE
- Samanantar dataset card: corpus statistics, domain breakdown, and download instructions.
  https://huggingface.co/datasets/ai4bharat/samanantar
- SacreBLEU documentation: detokenised BLEU and comparison to NIST variants.
  https://github.com/mjpost/sacrebleu
- Claude (Anthropic): LLM assistant used for code development, debugging of the training pipeline, and structuring of this project report.
  https://claude.ai

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